2. We read ∂F as **"partial F"** and ∂x as **"partial x"**. And we read $\frac{\partial F}{\partial x}$ as "partial F by partial x". It is also read as "dho F by dho x".

3. Similarly, we read $\frac{\partial F}{\partial x}$ as "partial F by partial y" or as "dho F by dho y".

4. Sometimes $\frac{\partial F}{\partial x}(x_0, y_0)$ is also denoted by $F_x(x_0, y_0)$ or $\frac{\partial F}{\partial x}(x, y)\Big|_{(x_0, y_0)}$

   Similarly $\frac{\partial F}{\partial y}(x_0, y_0)$ is denoted by $F_y(x_0, y_0)$, or $\frac{\partial F}{\partial y}(x, y)\Big|_{(x_0, y_0)}$

5. An important thing to notice is that while finding partial derivative of F with respect to x, we treat the y variable as a constant and find derivative with respect to x. That is, except for the variable with respect to which we find partial derivative, all other variables are treated as constants. That is why we call it as **"partial derivative"**.

6. If F has a partial derivative with respect to x at every point of A, then we say that $\frac{\partial F}{\partial x}(x, y)$ exists on A. Note that in this case $\frac{\partial F}{\partial x}(x, y)$ is again a real-valued function defined on A.

7. In the light of (4), it is easy to see that all the rules **(Sum, Product, Quotient, and Chain rules)** of differentiation and formulae that we have learnt earlier hold true for the partial differentiation also.

Recall that for a function of one variable, differentiability at a point always implies continuity at that point. For a function F of two variables x, y we have defined $\frac{\partial F}{\partial x}(u, v)$ and $\frac{\partial F}{\partial y}(u, v)$. Do the existence of partial derivatives of F at a point (u, v) implies continuity of F at (u, v)? Following example illustrates that this may not necessarily happen always.

Example 8.11

Let $f(x, y) = 0$ if $xy \neq 0$ and $f(x, y) = 1$ if $xy = 0$.

(i) Calculate : $\frac{\partial f}{\partial x}(0,0)$, $\frac{\partial f}{\partial y}(0,0)$.

(ii) Show that f is not continuous at $(0,0)$.

Solution

Note that the function f takes value 1 on the x, y-axes and 0 everywhere else on $\mathbb{R}^2$. So let us calculate

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h \to 0} \frac{f(0+h, 0) - f(0,0)}{h} = \lim_{h \to 0} \frac{1-1}{h} = 0;$$

$$\frac{\partial f}{\partial y}(0,0) = \lim_{k \to 0} \frac{f(0, 0+k) - f(0,0)}{k} = \lim_{k \to 0} \frac{1-1}{k} = 0.$$

This completes (i).

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